11.4.1 Relational Database Management System (RDBMS)

RDBMS is more advanced version of DBMS, that allows to data in a more efficient way. It is used to store and manupulate the data that are in the form of tables.

Examaple of RDBMS: MySQL,Oracle, MS-Access etc.,

11.4.2 Characteristics of Relational Database Management System

1. Ability to manipulate data	RDBMS provides the facility to manipulate data (store, modify and delete)in a data base.
2. Reduced Redundancy	In the modern world hard drives are very cheap, but earlier when hard drives were too expensive, unnecessary repetition of data in database was a big problem But RDBMS follows Normalisation which divides the data in such a way that repetition is minimum.
3.Data Consistency	On live data, it is being continuously updated and added, maintaining the consistency of data can become a challenge. But RDBMS handles it by itself.
4. Support Multiple user and Concurrent Access	RDBMS allows multiple users to work on it(update, insert, delete data) at the same time and still manages to maintain the data consistency.
5.Query Language	RDBMS provides users with a simple query language, using which data can be easily fetched, inserted, deleted and updated in a database.
6. Security	The RDBMS also takes care of the security of data, protecting the data from unauthorized access. In a typical RDBMS, we can create user accounts with different access permissions, using which we can easily secure our data by restricting user access.
7. DBMS Supports Transactions	It allows us to better handle and manage data integrity in real world applications where multi-threading is extensively used.

11.4.3 Advantages of RDBMS

- Segregation of application program
- Minimal data duplication or Data Redundancy
- Easy retrieval of data using the Query Language
- Reduced development time and maintenance

12th Computer Science_EM Chapter 11.indd 183 26-12-2022 16:46:12